

KudiReady: A Cloud-Native Credit-Readiness System for Nigerian MSMEs

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Abstract

KudiReady is a cloud-deployed web system that turns a Nigerian micro or small business's bank and mobile-money statements into an explainable, lender-ready credit file. It cleans and categorises an uploaded CSV statement, scores readiness across four 25-point pillars, lists what to fix first, and lets the owner share an expiring, read-only snapshot with a lender. The system is built with React and TypeScript on Vercel, a FastAPI service in Docker on Render, and PostgreSQL 16, deployed through a gated GitHub Actions pipeline; the backend has 39 tests at 94% coverage. On a synthetic 12-month statement, fixing the gaps the system identified raised the readiness score from 68 to 76.

Live system: kudiready.vercel.app · **API documentation:** kudiready-api.onrender.com/docs · **Source code:** github.com/ADAGBON/kudiready

Author's note: this is a solo project, designed and built independently of my earlier team capstone (Onrecord, Ghana-focused). KudiReady targets Nigeria and shares no code with it.

1. Introduction and problem statement

Nigerian micro, small and medium enterprises (MSMEs) are short of credit not mainly because they are bad risks, but because their financial evidence is not in a form lenders can read. This project builds a tool that closes that legibility gap.

The scale of the sector is large. The SMEDAN–NBS national survey counted about 39.6 million MSMEs in 2020, and SMEDAN reports their contribution to GDP at 46.31% ([Vanguard, 2023](#)). The earlier 2013 survey found MSMEs employed 84% of the labour force, and that 96% of micro enterprises were not registered ([NBS & SMEDAN, 2015](#)).

Finance is the sector's binding constraint. In PwC's 2024 Nigeria MSME survey, inadequate access to finance was the most-cited growth challenge, named by 35% of firms; family and friends (38%) were a more common funding source than bank loans and overdrafts (26%). PwC cites an IFC estimate of the unmet MSME credit need at about \$32.2 billion ([PwC, 2024](#)).

Working at the Central Bank of Nigeria for five years, across treasury reconciliation and monetary statistics, I saw the other side of this gap. A market trader's money moves daily through POS terminals, OPay and bank transfers, so the evidence exists — but as 1,500 unlabelled statement lines. The owner learns what is missing only when turned down.

Problem statement. There is no simple, free tool that lets a Nigerian MSME owner see their own statement the way a lender will, understand what is weakening it, and fix it before applying.

Objective. Design, build and deploy a production-grade, cloud-native web system that (1) ingests real-world Nigerian statement exports, (2) produces a transparent readiness assessment with a prioritised fix list, and (3) lets the owner share a verifiable snapshot with a lender — while protecting the owner's financial data.

Scope boundary. KudiReady does not lend, does not produce a credit score in the regulatory sense, and does not make credit decisions. It measures how complete, consistent and lender-legible a business's records are.

2. Literature review

The literature points to information, not only risk, as the core of the MSME credit gap, and to transaction data as the most practical way to close it.

2.1 Information asymmetry in small-business lending

Stiglitz and Weiss (1981) showed that when lenders cannot observe borrower quality, they ration credit rather than raise prices, because higher rates drive out safer borrowers. Informal firms, with no audited accounts or bureau history, are the extreme case, so lenders substitute collateral for information. PwC's 2024 survey shows the consequence from the borrower's side: among firms that applied for loans, high interest rates (27%), lengthy procedures (26%), insufficient collateral (16%) and inadequate documentation (14%) were the leading obstacles ([PwC, 2024](#)). Procedures and documentation together outrank collateral — that is, much of the friction is about producing evidence, which is a solvable software problem.

2.2 Cash-flow data as an alternative to collateral and bureau history

A growing body of evidence shows that bank-account transaction data predicts repayment for thin-file borrowers. FinRegLab's 2019 study of six non-bank lenders found that cash-flow variables and scores derived from account records were predictive of

credit risk and loan performance, and could extend access to consumers and small firms that traditional scores miss ([Flourish Ventures summary of FinRegLab, 2019](#); [FinRegLab](#)). KudiReady's indicators — average inflow, volatility, cash-positive months, existing repayment burden — are the ones a credit officer already reads by hand.

2.3 The Nigerian context: digital rails and open banking

Two features of Nigeria make this approach unusually feasible. First, small-business money already moves through digital rails — POS agents, NIP instant transfers, and fintech wallets such as OPay, Moniepoint and PalmPay — so most traders can export a statement today. Second, the Central Bank issued Operational Guidelines for Open Banking on 7 March 2023, requiring authenticated customer consent and JSON-based APIs for sharing account data ([Afriwise, 2023](#)). That creates a path from user-uploaded CSVs today to bank-verified data feeds later, without redesigning the system.

2.4 Data protection obligations

The Nigeria Data Protection Act, signed on 14 June 2023, established the Nigeria Data Protection Commission and requires processing to follow lawfulness, purpose limitation, data minimisation, storage limitation and security principles ([Mondaq, 2023](#)). Statements are sensitive financial data, so these principles shaped concrete design choices (Section 6).

2.5 Gap this project addresses

Existing tools serve lenders; few show the borrower how their file reads and what to fix. KudiReady sits there. It deliberately uses transparent, rule-based scoring rather than a trained model, for two reasons supported by the literature: there is no public Nigerian MSME repayment dataset to train on, and an owner can only act on a score they can understand.

3. Requirements and design goals

The requirements come from two users: the owner, usually on a phone, who wants to know where they stand and what to do; and the loan officer, who wants a complete, consistent file without creating an account.

ID	Requirement	Type	How it is met
F1	Owners register, log in and keep one business profile	Functional	bcrypt + JWT auth; one-to-one user-business
F2	Import real Nigerian statement CSVs without manual reformatting	Functional	Header-alias matching, signed or debit/credit layouts, 9 date formats, ₦/NGN/brackets
F3	Re-uploading overlapping statements never double-counts money	Functional	Per-line SHA-256 fingerprint + unique constraint
F4	Every line categorised; owner can override	Functional	Ordered keyword rules; PATCH endpoint
F5	Explainable score plus a prioritised fix list	Functional	Deterministic engine; per-component points and reasons
F6	Share a read-only, expiring, revocable view with a lender	Functional	Hashed tokens, frozen snapshot, 14-day expiry
N1	One owner can never see another's data	Security	Business resolved from the token, never from the URL
N2	Minimal personal data stored	Privacy	Yes/no flags instead of CAC, TIN or BVN numbers
N3	Usable on a 375 px phone	Usability	Responsive layout, bottom navigation
N4	Exact, reproducible money arithmetic	Correctness	Integer kobo, deterministic scoring
N5	Every change tested and deployed automatically	Maintainability	GitHub Actions gate before Render and Vercel deploy
N6	Scales horizontally without code changes	Scalability	Stateless API, managed Postgres, CDN-served frontend

Two design principles governed trade-offs. **Explain before predict:** every point in the score must trace to a number the owner can see and change. **Fail per row, not per file:** a statement with three malformed lines out of 1,700 should import 1,697 and report the three.

4. System architecture

KudiReady is a three-tier system: a static single-page app on a global CDN, a stateless Python API in a container, and a managed relational database. Each tier scales and fails independently.

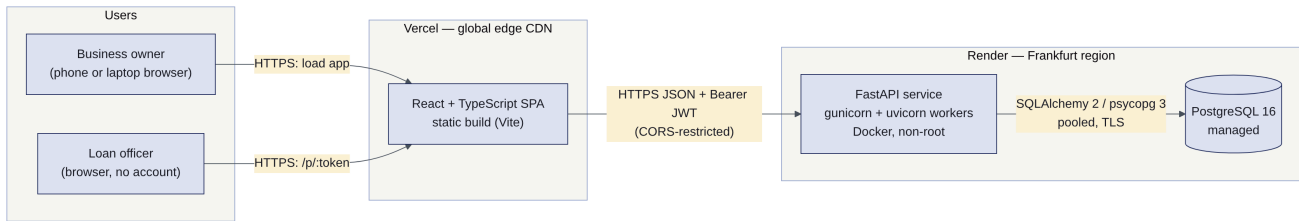


Figure 1. System architecture. Owners and lenders load the React app from Vercel's edge; it calls the FastAPI service on Render over HTTPS with a bearer token; the API is the only component that touches Postgres.

4.1 Technology choices

Layer	Choice	Why this over the alternatives
Frontend	React 19, TypeScript, Vite	Typed API contracts; 93 KB gzipped bundle for mobile data. Next.js rejected: nothing needs server rendering.
Frontend hosting	Vercel	Edge CDN, automatic HTTPS, per-PR previews.
API	FastAPI + Pydantic v2	Validation and OpenAPI docs from one set of type hints. Django was heavier than needed.
ORM and migrations	SQLAlchemy 2 + Alembic	Explicit SQL control; versioned migrations.
Database	PostgreSQL 16	Ledgers need ACID transactions and unique constraints; a document store would push integrity into app code.
API hosting	Render (Docker)	One image locally, in CI and in production; infrastructure as code.
Server	gunicorn managing uvicorn workers	Process supervision, multi-core use.

Synchronous database sessions were chosen over async drivers: requests are short indexed queries plus arithmetic, so async would add complexity without gain.

4.2 Backend structure

The backend separates HTTP concerns from domain logic. Routers handle authentication, validation and persistence; the CSV parser, categoriser and readiness engine are pure functions with no database or network access, which makes them fast and exhaustively testable.

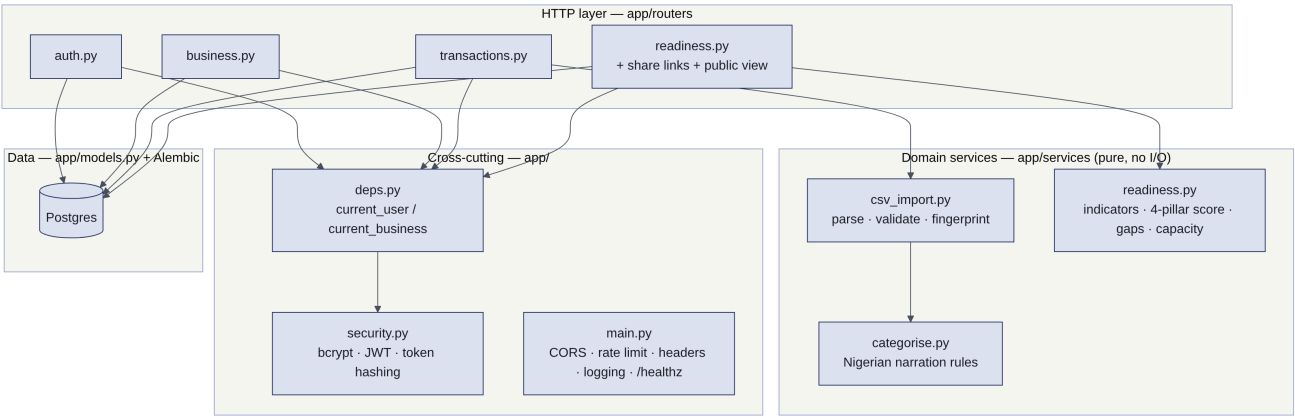
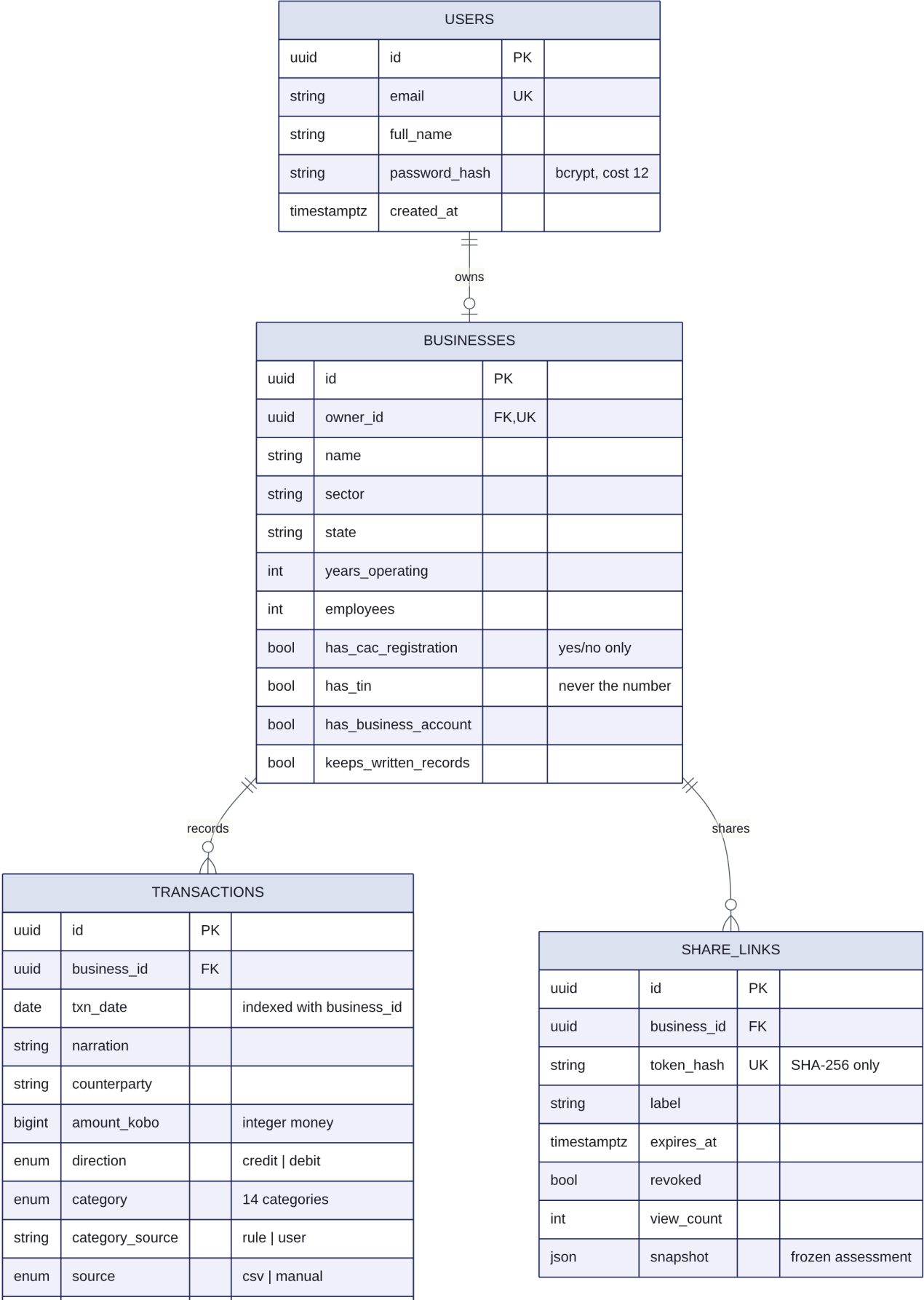


Figure 2. Backend modules. Domain services in app/services have no I/O and are unit-tested directly.

4.3 Data model



string	fingerprint		UK with business_id
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Figure 3. Four tables. Money is BIGINT kobo; the (business_id, fingerprint) unique constraint enforces idempotent imports; share links store only a SHA-256 hash of the token.

4.4 Request flow

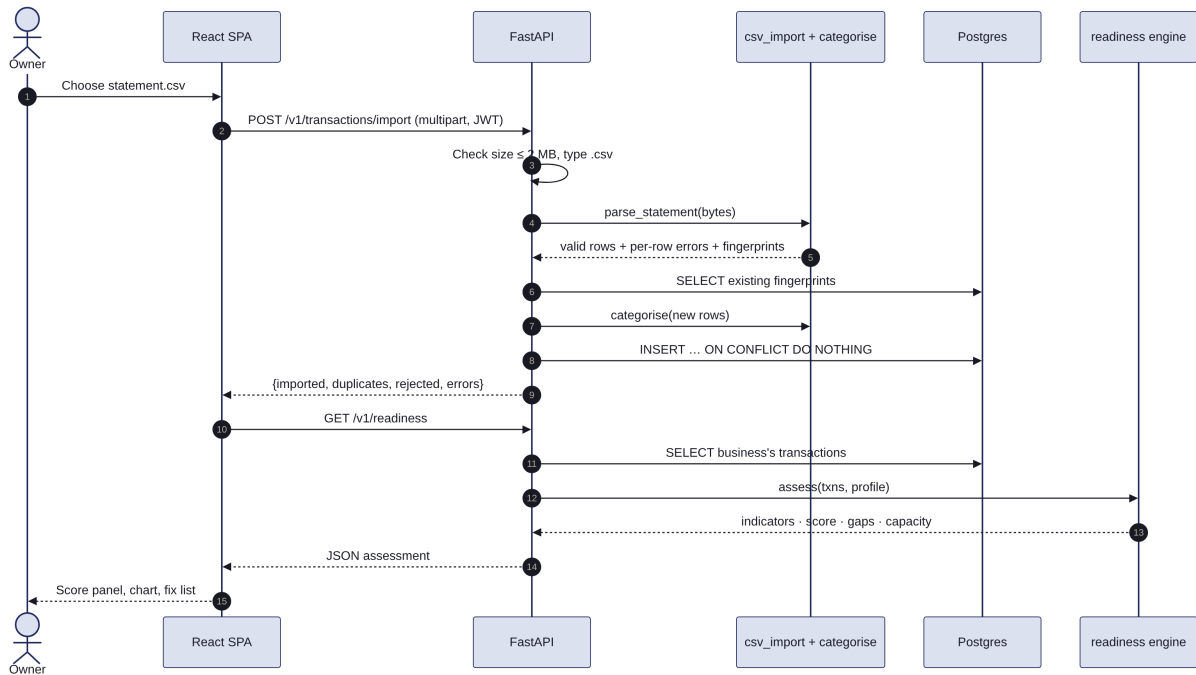


Figure 4. An import is validated, parsed per row, de-duplicated against existing fingerprints, categorised and inserted in one transaction; the assessment is recomputed on read.

Scores are computed on read, not stored: the engine takes about 13 ms on 1,800 lines, and no cached score can go stale.

5. Implementation

The implementation is about 1,450 lines of application Python (834 executable statements) and 1,000 lines of TypeScript, organised so each module has one job. Four parts carry the system's value.

5.1 Statement import

The parser matches headers against alias lists (*narration, description, details, remarks*), accepts signed-amount or debit/credit layouts, strips ₦ and NGN, and tries nine date formats such as *04-Mar-2026*. Rows are validated independently, so one bad line is reported, not fatal. Uploads are capped at 2 MB and 5,000 rows.

Each line gets a SHA-256 fingerprint of date, narration, amount, direction, reference and an occurrence counter — so two identical same-day POS sales are both kept, but a re-uploaded statement adds nothing. A unique constraint with ON CONFLICT DO NOTHING makes concurrent duplicates safe too.

5.2 Categorisation

An ordered set of keyword rules maps narrations to 14 categories, separately for money in and money out. Rules encode local vocabulary: *POS, NIP TRF FROM, Moniepoint* and *OPay* credits indicate sales; *LAPO, FairMoney* and *Carbon* debits indicate loan repayments; *IKEDC, airtime* and *diesel* indicate utilities; *FIRS, LIRS* and *LASG levy* indicate tax. Own-account transfers and loan disbursements are excluded so they cannot inflate revenue. Unmatched lines become a gap the owner fixes in one click.

5.3 Readiness engine

The engine scores the last 12 months across four 25-point pillars mirroring an MSME credit officer's checks:

Pillar	Components (points)
Record quality	Months of records (12), continuity (6), categorised share (4), CAC/TIN/business account (3)
Cash-flow strength	Operating margin (13), share of cash-positive months (12)
Stability and trend	Inflow volatility as coefficient of variation (15), last-3 vs first-3-month growth (10)
Obligations and risk	Existing debt-service ratio (13), largest-customer share (8), years operating (4)

Each component is a linear scale between a floor and a target (for example, margin earns nothing at 0% and full points at 30%), so every point is traceable. Bands are Not ready (under 40), Building (40–64), Nearly ready (65–79) and Lender-ready (80+).

Testing exposed a flaw in this first design: two months of strong data scored 67, "Nearly ready". No lender treats two good months as evidence of repayment capacity. I added an explicit **evidence cap** — the score cannot exceed 35 with fewer than three months of records, or 55 with fewer than six — and the cap reason is shown to the owner.

Weak components become actions ranked by severity and recoverable points. Indicative capacity — 35% of monthly free cash after costs and existing repayments — is labelled a planning aid, not an offer.

5.4 Frontend

Five owner screens and one public lender view. Each pillar renders as 25 tally marks, borrowed from traders' paper ledgers; the hand-built SVG chart marks missing months. Copy is plain English: *Kept after running costs, not operating margin*.

6. Security, privacy and scalability

The most important control, tenant isolation, is structural rather than a check that could be forgotten.

6.1 Security controls

- **Tenant isolation.** The business is always resolved from the authenticated user, never from a client-supplied ID; another owner's record returns 404. A test attacks this as a second user.
- **Credentials.** bcrypt at cost 12; unknown emails are checked against a dummy hash so timing does not reveal registered accounts.
- **Tokens.** Expiring HS256 JWTs; the API refuses to start in production with a weak or default secret.
- **Share links.** 192-bit tokens stored only as SHA-256 hashes; 14-day expiry; revocable.
- **Abuse limits.** 10 auth attempts per IP per minute; capped, type-checked uploads.
- **Transport and headers.** HTTPS throughout, CORS locked to the frontend origin, security headers.
- **Container.** Slim multi-stage image running as a non-root user.

6.2 Privacy by design (NDPA 2023)

Data minimisation is applied literally: the system asks whether the owner has a CAC registration, TIN and business account, and stores only yes or no — never the identifiers, and never a BVN. Purpose limitation shapes sharing: a named recipient, a limited time, a visible view count. Owners can delete their data.

6.3 Scalability

The stateless API scales horizontally behind Render's load balancer; the static frontend is CDN-served; the main query uses a composite (business_id, txn_date) index with pooled connections.

At larger scale, the in-process rate limiter moves to Redis and big imports to a background worker; today 1,700 lines import in about 170 ms.

7. Testing, CI/CD and cloud deployment

No code reaches production without passing three automated gates, and the production database schema is only ever changed by versioned migrations.

7.1 Test strategy

The backend has 39 automated tests reaching 94% line coverage, in three layers:

Layer	Tests	What they prove
Readiness engine (unit)	11	Indicator arithmetic, 12-month windowing, missing-month detection, exclusion of transfers and loan inflows, evidence cap, debt and capacity maths, determinism, categorisation rules
CSV parser (unit)	16	Five date formats, six money formats, both statement layouts, per-row errors, identical same-day lines, missing columns, row limits
API (integration)	12	Register/login/me, duplicate email, weak password, auth required, business validation, full import → score → fix → share → revoke flow, re-upload idempotency, tenant isolation, manual entry, bad files, rate limiting

Tests run on in-memory SQLite locally (under six seconds) and on real PostgreSQL 16 in CI, where Postgres-specific behaviour is exercised. The suite caught two real defects: over-generous scoring of short histories, and identical same-day sales being discarded as duplicates.

An automated Playwright walkthrough verified the owner and lender journeys at desktop and mobile widths; Section 8's screenshots are its output.

7.2 Continuous integration and delivery

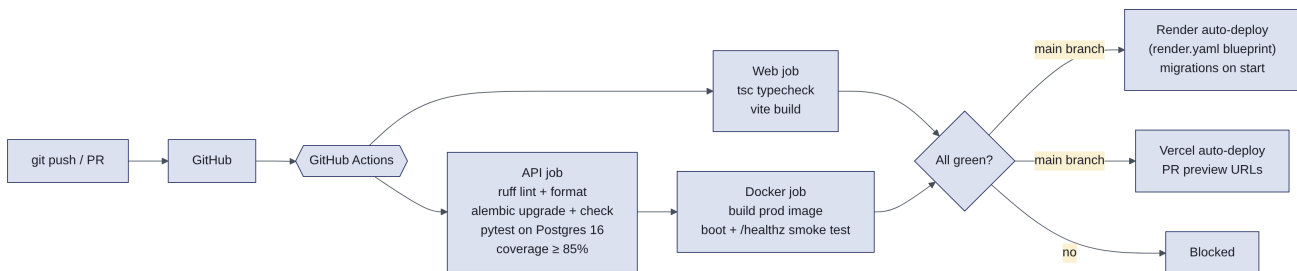


Figure 5. Every push and pull request runs three jobs in GitHub Actions; deploys follow only from a green main branch.

1. **API job:** ruff lint; alembic upgrade head on Postgres 16, then alembic check to catch model-migration drift; pytest with an 85% coverage floor.
2. **Web job:** TypeScript strict type-check and production Vite build.
3. **Docker job:** builds and boots the production image with ENVIRONMENT=production, polling /healthz until healthy.

7.3 Deployment

A Render Blueprint (render.yaml) declares the Docker API and managed PostgreSQL 16 in Frankfurt, the closest region to Lagos, injecting the database URL and generating the JWT secret. Migrations run on container start. The health check uses /healthz (including a database check), and deploys wait for CI. Vercel builds the frontend from the same repository.

The system is publicly accessible at kudiready.vercel.app, with interactive API documentation at kudiready-api.onrender.com/docs.

8. Results and evaluation

The system meets all twelve requirements in Section 3, and the scenario shows the intended loop: it finds real weaknesses, and fixing them measurably improves the file.

8.1 Evaluation scenario

A synthetic 12-month statement for a Lagos provisions shop (1,828 lines: sales, restocking, rent, pay, an MFI loan, utilities, levies, December seasonality) was split so February is "lost".

Step	Result
Upload main statement (1,695 lines)	All imported in ~170 ms, 0 rejected; 22 lines left uncategorised
Upload the same file again	0 imported, 1,695 skipped as duplicates
First assessment	68/100, Nearly ready; February flagged; weakest pillar cash-flow strength (11% margin)
Upload February, categorise 22 lines, add CAC and TIN	76/100; Record quality rises from 21 to 25 of 25
Share with lender, open, revoke	Snapshot served without login; view counted; link returns "unavailable" after revoke

The score did not move into Lender-ready, and that is the right answer: the remaining weakness is an 11% operating margin, a business fundamental that better paperwork cannot fix.

K

KudiReady

Overview

Transactions

What to fix

Share

Profile

Adaeze Okafor

Log out

Mama Ada Provisions

11 months of records (Oct 2025 to Sep 2026), average monthly inflow ₦1,397,822. Score 68/100 (Nearly ready); weakest area: cash-flow strength.

Share with a lender

68 / 100

Nearly ready

Record quality

21/25

Cash-flow strength

10/25

Stability & trend

17/25

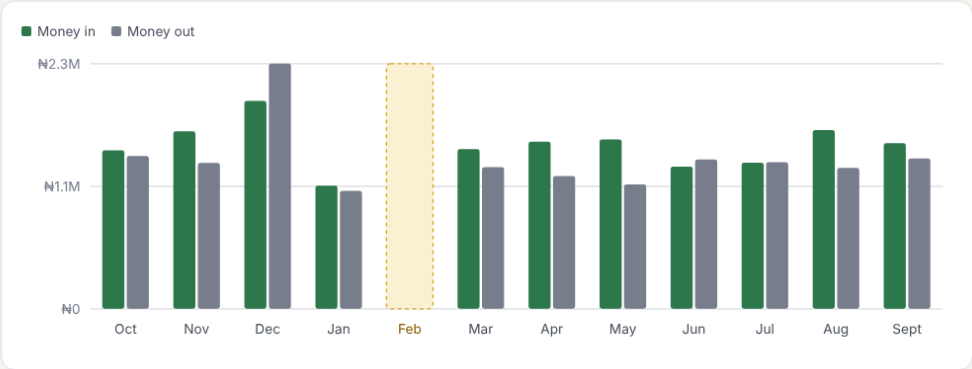
Obligations & risk

20/25

₦1.4M Average money in per month	11% Kept after running costs	11 months October 2025 – September 2026	₦40.4K Repayment you could carry monthly*
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* Indicative only: 35% of your average monthly free cash after all costs and existing loan repayments. It is a planning aid, not a loan offer or credit decision.

Money in and out by month



- What to fix first

See all 6
- Register your business with CAC

A CAC business-name registration (Business Name, ~₦10k–₦20k) is required by most bank SME loans.

Registration

up to +1
- Extend your records to 12 months

You have 11 months. Twelve months shows lenders a full trading cycle, including seasonality.

Records

up to +1
- Get a Tax Identification Number (TIN)

Free from FIRS / JTB. Many lenders and government schemes require a TIN.

Registration

up to +1

How your score is made

Record quality	21.3 / 25	Cash-flow strength	10.0 / 25
Months of records 11 of 12 months covered	10.9 / 12	Operating margin 11% of inflows retained	4.7 / 13
Continuity 1 missing month(s)	5.5 / 6	Cash-positive months 67% of months inflow > outflow	5.3 / 12
Categorised transactions 99% categorised	3.9 / 4		
Formal registration 1 of 3 (CAC, TIN, business account)	1.0 / 3		

Stability & trend	16.6 / 25	Obligations & risk	20.2 / 25
Revenue stability	12.6 / 15	Existing debt burden	13.0 / 13
Monthly inflow varies ±33%		3% of inflows go to loans	
Revenue trend	4.0 / 10	Customer concentration	4.0 / 8
-10% last 3 vs first 3 months		Counterparties not recorded	
		Years operating	3.2 / 4
		4 year(s)	

Figure 6. Overview at 68/100. The missing February is highlighted in the chart; the fix list leads with CAC registration and missing months; every score component is explained below.

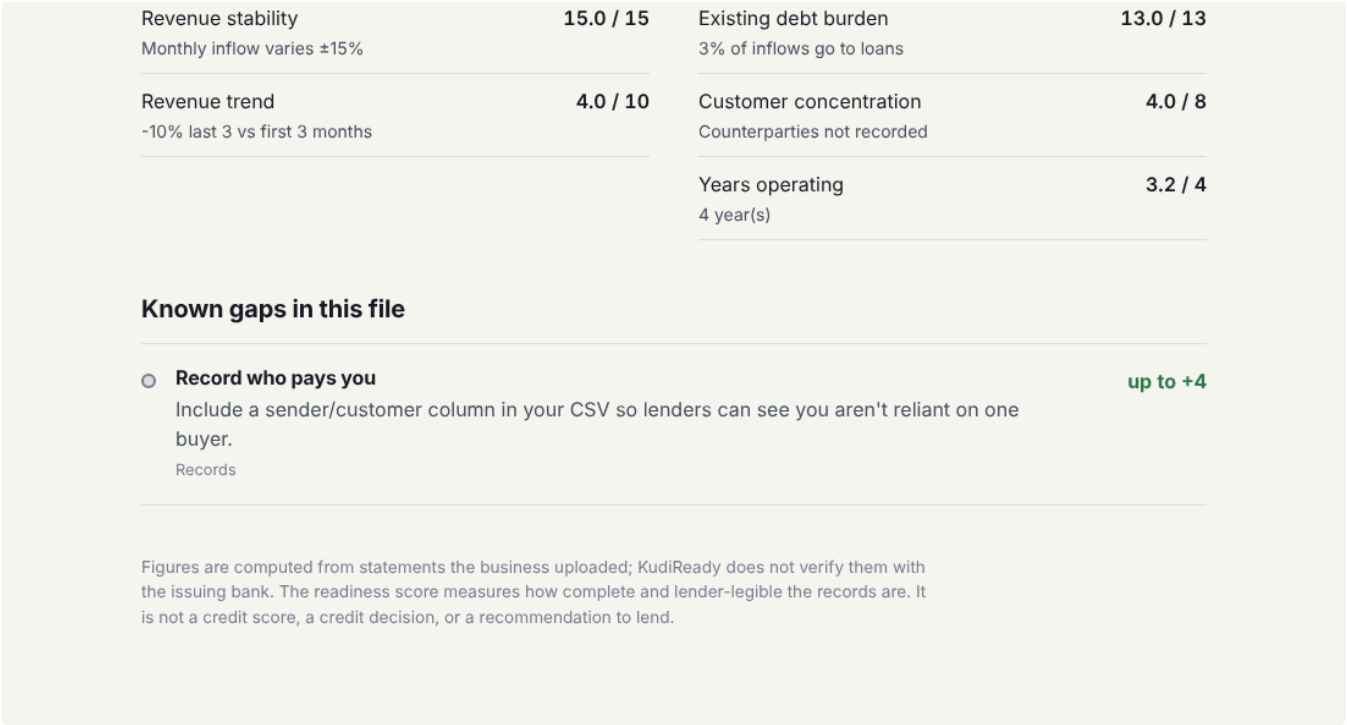


Figure 7. The lender’s read-only view at 76/100, with the business profile, recipient, date and a disclaimer that this is not a credit decision.

Mama Ada Provisions

12 months of records (Oct 2025 to Sep 2026), average monthly inflow ₦1,491,912. Score 76/100 (Nearly ready); weakest area: cash-flow strength.

Share with a lender



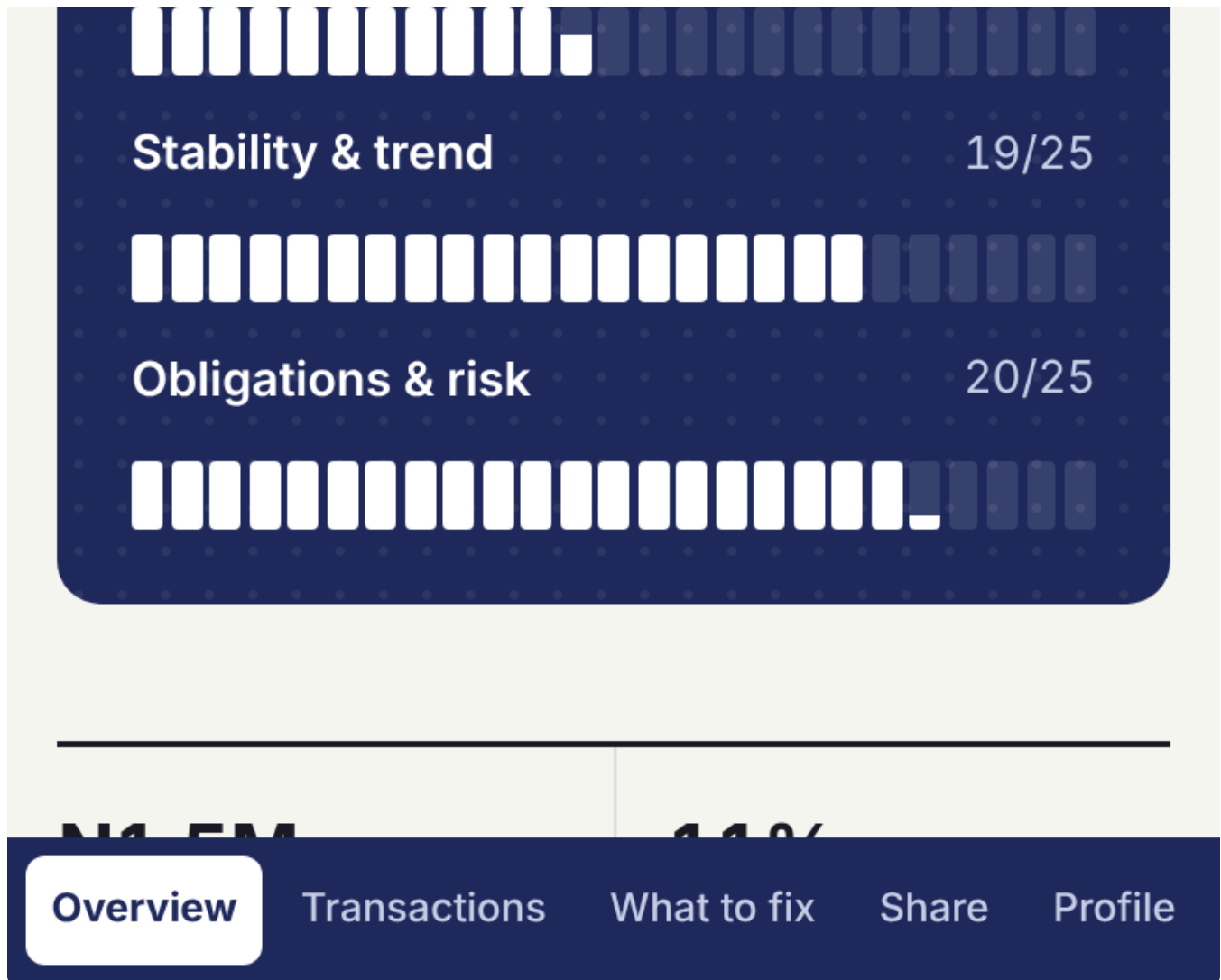


Figure 8. The same overview at 390 px, the width of a typical Android phone.

8.2 Evaluation against objectives

- **Real-world data handling:** a realistic export loaded without reformatting.
- **Explainability:** every point has a visible reason.
- **Production readiness:** containerised, migrated, health-checked, 94% coverage, gated deploys.
- **Performance:** about 170 ms to import 1,700 lines, 13 ms to assess.

9. Limitations, future work and conclusion

9.1 Limitations

- **Self-reported data.** Uploads are not verified with the bank; an edited CSV would be accepted. The lender view says so.
- **Uncalibrated weights.** Weights reflect lending practice and my CBN experience, not repayment outcomes; no public Nigerian MSME loan dataset exists.
- **Rule-based categorisation.** Unseen narrations need manual correction.
- **CSV only.** PDF statements are not yet parsed.
- **Free-tier hosting.** The idle API sleeps, so the first request can take 30–50 seconds.

9.2 Future work

1. **Verified data via open banking.** Consent-based account access under the CBN framework, removing self-reporting.
2. **Outcome calibration.** Fit weights to repayment data with a microfinance partner.
3. **PDF statement parsing** for the major banks and fintech wallets.
4. **Learning categoriser.** Learn from owners' corrections, keeping rules as the baseline.
5. **Operational scale-out.** Redis rate limiting, background imports, metrics and alerting.

9.3 Conclusion

KudiReady shows that a large part of the MSME credit gap can be addressed without lending a naira: by turning the evidence small businesses already generate into a file lenders can read, and by telling owners, specifically, what to fix. It is a working cloud-native application built around explicit decisions on money handling, idempotency, tenant isolation and data minimisation. Its most useful result was the one testing forced: a readiness tool must refuse to be optimistic about thin evidence, just as a good credit officer would.

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Appendix — repository contents: `backend/` (FastAPI app, Alembic migrations, 39 tests, Dockerfile) · `frontend/` (React app, vercel.json) · `docs/` (architecture diagrams, screenshots, deployment guide, demo script) · `render.yaml` · `.github/workflows/ci.yml`.